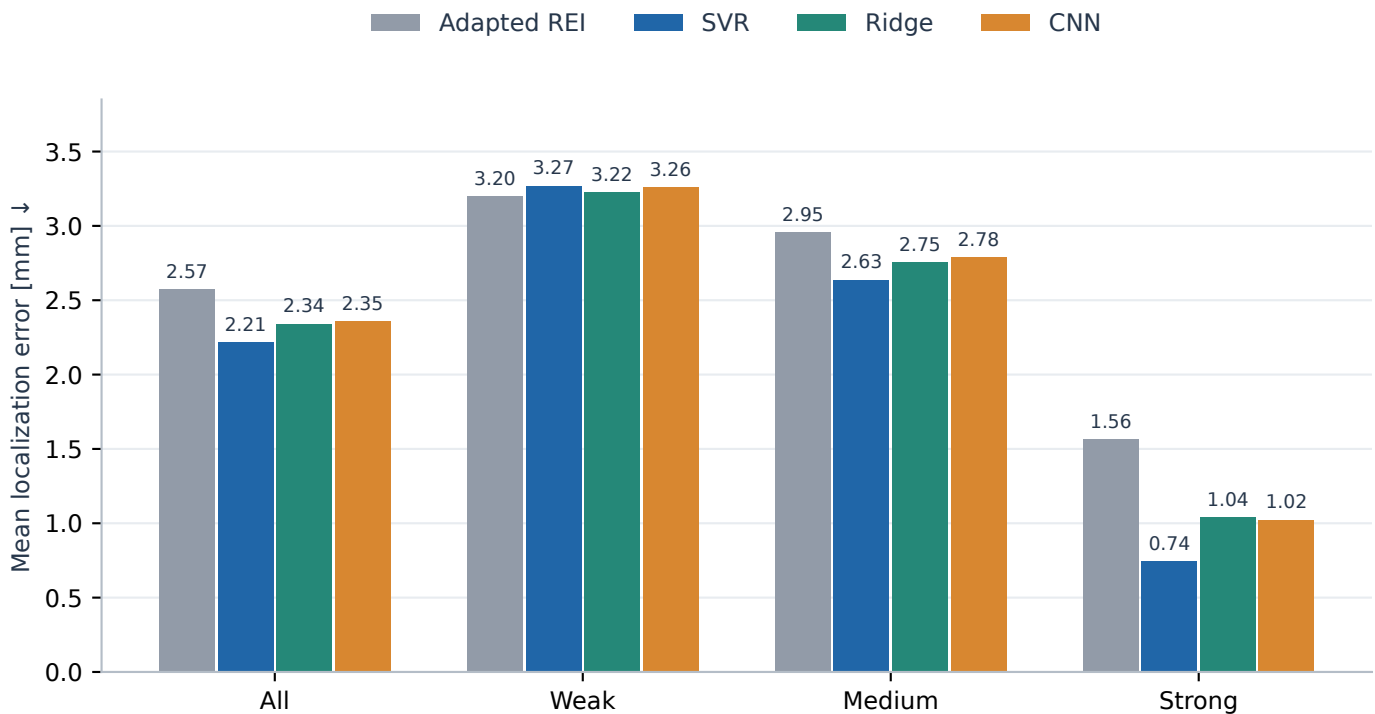


# Localizing radiation events

Primary result: SVR lowers overall mean error versus adapted REI using the same records.

Input: binary surface-code detector events → Output: epicenter coordinates (x, y).

The window contains one event. Event detection and onset-time estimation are not evaluated.



Overall mean: 2.571 → 2.214 mm (13.9% lower). SVR–REI, 95% CI: –0.414 to –0.301 mm.

Strong events within 1 mm: 23.9% → 77.8%. Medium: only 7.5%; weak events did not improve.

## Matched evaluation

d5 rotated surface-code memory-Z: 49 qubits, 24 checks, 1  $\mu$ s per round.

Train / validation / nominal test: 864 / 180 / 540 physical events; two shots each, 180 tests per band.

All methods use the same 4,095 internal rounds (~4 ms); settings are frozen before test generation.

Only coordinates supervise learning. CNN averages three initializations; search budgets differ.

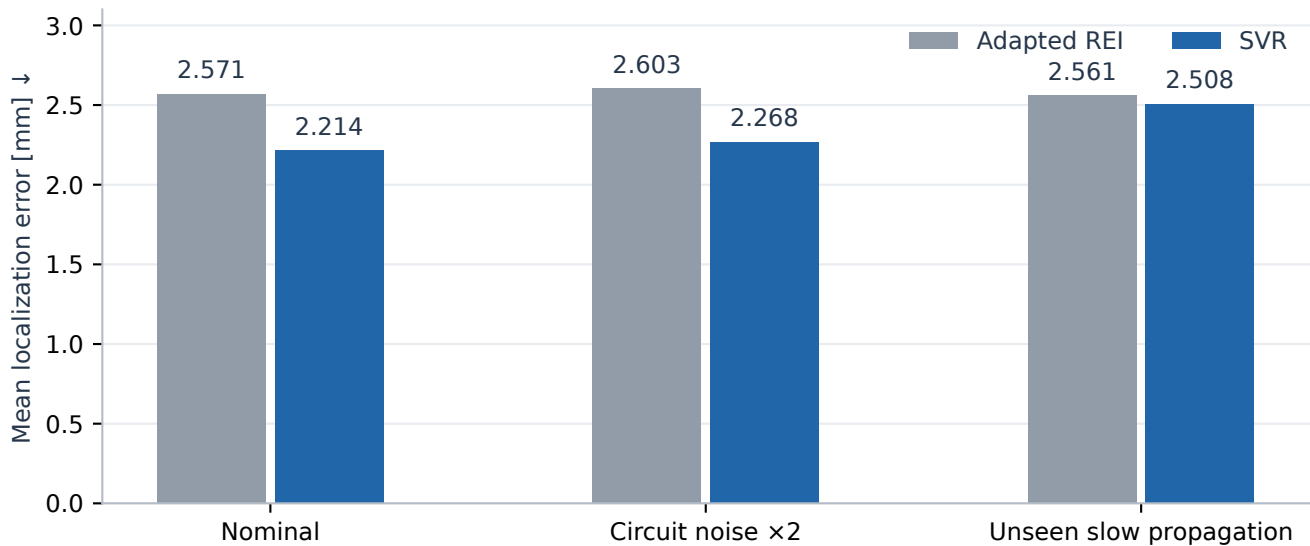
Error is Euclidean coordinate distance. The overall mean gives equal weight to the three bands.

Overall mean is the prespecified primary contrast; paired event bootstrap uses 10,000 draws.

[1] rei\_fair\_20260911 | Figure: group means; CI refers to the primary paired difference.

# Where the improvement holds

Additional analyses from the matched study. All contrasts except the primary are exploratory.



The mean benefit persists with doubled circuit noise. Under unseen slow propagation, SVR–REI is  $-0.053$  mm (95% CI  $-0.147$  to  $+0.042$ ): no clear advantage. CNN: 2.452 mm (point estimate).

## Strong events: improvement for both shapes

Circular: REI 1.556  $\rightarrow$  SVR 0.653 mm. Elliptical: 1.568  $\rightarrow$  0.832 mm.

Both shapes are in training; this is not evidence of generalization to an unseen shape.

## Lower mean error does not imply a better tail

Nominal  $\sim 4$  ms overall p90 (90th error percentile): REI 4.071  $\rightarrow$  SVR 4.210 mm, worse.

Weak-event mean: 3.198  $\rightarrow$  3.266 mm, with no improvement. These cases remain in the report.

Tests: 540 nominal events and their noise-shifted versions, plus 216 independent slow events;

756 unique physical events. Only gate, measurement and reset noise are doubled, not T1/T2.

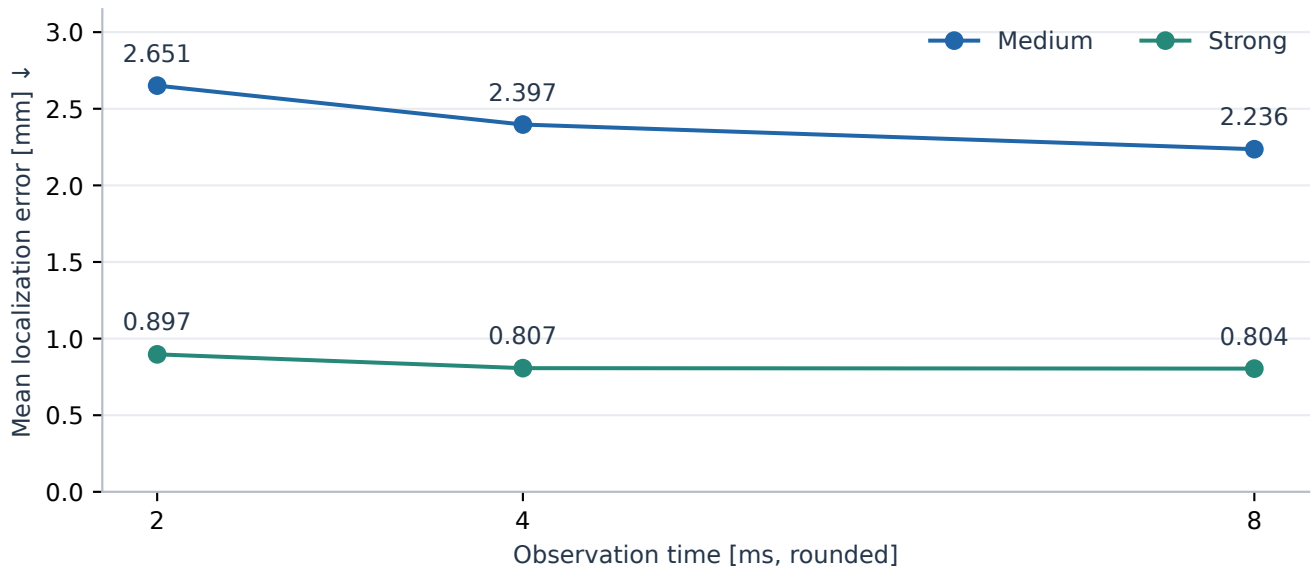
Slow speed: 2–6 vs training 12–40 m/s; diffusion: 0.5–2 vs training 5–20 mm<sup>2</sup>/ms.

Models and nominal calibration stay fixed. Later arrival within the fixed window also affects results.

[1] rei\_fair\_20260911 | All results on this page use  $\sim 4$  ms.

# Observation time and feature choice

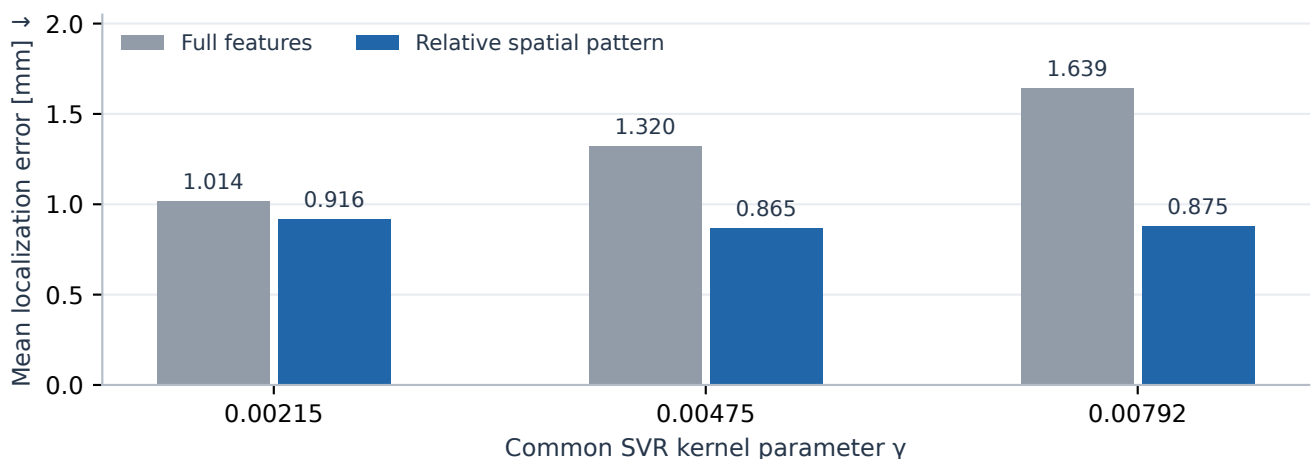
Supporting studies: matched tests within each panel; different training/tests from pages 1-2.



Medium events improve up to ~8 ms; the strong-event 4→8 ms gain is unclear.

Medium, 2→8 ms:  $-0.415$  mm (98.75% CI  $-0.541$  to  $-0.294$ ; adjusted for four primary contrasts).

Same prefixes; 1,080 train / 540 test events. Models are refit; feature dimension and  $\gamma$  vary by window.



Representation: relative spatial patterns give lower strong-event means at all three shared  $\gamma$  values.

Changing  $\gamma$  alone does not explain the benefit; feature dimension and distance distributions still differ.

This feature contrast is exploratory. Both supporting studies exclude strong ellipses from training.

# A focused research story

Focus: localization from syndrome records, its observation requirements, and its limits.

## Proposed central claim

On this simulation distribution, SVR lowers overall mean localization error versus adapted REI using the same records. Exploratory analyses show medium/strong gains. Weak events do not improve; the slow-shift advantage is unclear and nominal p90 worsens.

## Fairness and interpretation

1. Inference uses no test truth. Strength groups use true labels, not a tested strength classifier.
2. SVR/Ridge subtract quiet rates; CNN uses training-set scaling. Prior information differs from REI.
3. REI validation selects full history at  $\sim 4/8$  ms. Uncalibrated SVR: 2.257 mm at nominal  $\sim 4$  ms.
4. Adapted REI is not a full reproduction. Original-paper and computational superiority are unproven.
5. Intervals condition on training data; secondary and research-wide repeated searches are unadjusted.
6. One simulated device: phenomenological propagation + local QP-ODE + Pauli approximation.

## Keep separate from the primary comparison

Physical-template methods use latent probabilities during training; oracle results are diagnostics. Older data-size and code-distance studies can be supplementary, not a cross-test best-score leaderboard.

Weak / medium / strong are simulator generation bands, not universal measured-energy classes.

## Before submission

Repeat with independent training sets; validate physics and literature novelty; measure end-to-end latency.

Real hardware, BB codes, continuous monitoring, multiple events and decoding benefits are not established.

Sources: saved evaluation data and RESULT\_JA.md in each experiment directory.

[1] rei\_fair\_20260911 [2] long\_observation\_20260910 [3] feature\_selection\_20260910

Input SHA-256 hashes are in sources.json. Charts are generated from CSV; bars start at zero.